

Algebra 1
Unit 12
Lesson 1 Practice

Name _____
Date _____
Period _____

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Stem-and-leaf plots are useful to show large quantities of data. *True or False*
2. Line plots can show the range of small quantities of data. *True or False*
3. Box plots show overall shape for large quantities of data. *True or False*
4. A survey of 20 high school students was conducted to determine average time in minutes to travel to school. Which graphical displays would best display the data? *Select all that apply.*

- ☐ Box plot
- ☐ Line plot
- ☐ Histogram
- ☐ Stem-and-leaf plot

5. Explain your reasoning for question 4.
6. The data of the 20 high school students are below.

{20, 8, 12, 31, 9, 9, 11, 4, 23, 17, 16, 10, 10, 7, 21, 19, 7, 18, 15, 10}

Explain which display would be better for the data, a stem-and-leaf plot or a histogram.

Appliances-R-Us was analyzing their inventory before their next sale. The frequency table shows which appliances are in stock.

Frequency Table

Type of Appliance	Frequency
Microwave	22
Dishwasher	18
Oven	27
Refrigerator	13

Relative Frequency Table

Type of Appliance	Frequency
Rap	27.5%
Hip hop	22.5%
Rock	33.75%
Other	16.25%

7. In order to create a bar graph, the data from the

☐ relative frequency
☐ frequency

 table is needed.

8. In order to create a circle graph, the data from the

☐ relative frequency
☐ frequency

 table is needed.

9. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☐ Line Plot
☐ Circle Graph
☐ Stem-and-leaf plot
☐ Frequency Table
☐ Box plot
☐ Bar Graph
☐ Histogram

10. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☐ Line Plot
☐ Circle Graph
☐ Stem-and-leaf plot
☐ Frequency Table
☐ Box plot
☐ Bar Graph
☐ Histogram

Algebra 1
Unit 12
Lesson 2 Practice

Name _____
Date _____
Period _____

Identify whether each type of data is categorical or numerical. Then identify if it is univariate or bivariate.

1. The number of visitors to the beach and the average daily temperature.

- ☐ Categorical
- ☐ Numerical

- ☐ Univariate
- ☐ Bivariate

2. Whether it is sunny, partially cloudy, cloudy, or raining at the beach.

- ☐ Categorical
- ☐ Numerical

- ☐ Univariate
- ☐ Bivariate

3. The number of visitors to the beach on a summer day.

- ☐ Categorical
- ☐ Numerical

- ☐ Univariate
- ☐ Bivariate

Use the following for questions 4-6.

A local high school wanted to collect data on the number of absences of students for the 2020-2021 school year. The data of a sample of 20 high school students is shown.

{11, 2, 15, 8, 8, 9, 20, 35, 1, 4, 23, 23, 7, 7, 6, 15, 3, 4, 1, 9}

4. This data set is ☐ categorical ☐ numerical and ☐ univariate. ☐ bivariate.

5. Which is the best graphical representation to display the data? Choose one.

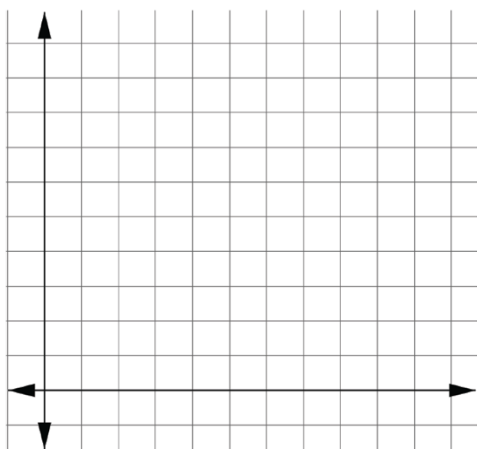
- ☐ Scatterplot
- ☐ Box plot
- ☐ Line Plot
- ☐ Histogram
- ☐ Circle Graph
- ☐ Stem-and-leaf plot
- ☐ Bar Graph

6. Create a graphical representation for the data set display chosen in question 5.

The number of people in a household and the weekly grocery costs are shown in the table below of 10 different families.

Number of People in the Household	Weekly Grocery Cost (\$)
2	90
6	210
3	125
2	105
5	150
8	225
4	130
4	100
5	180
7	200

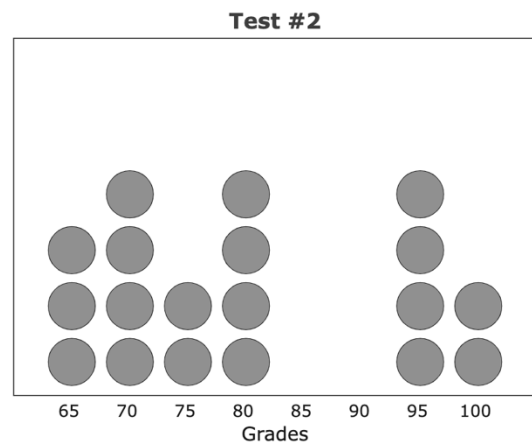
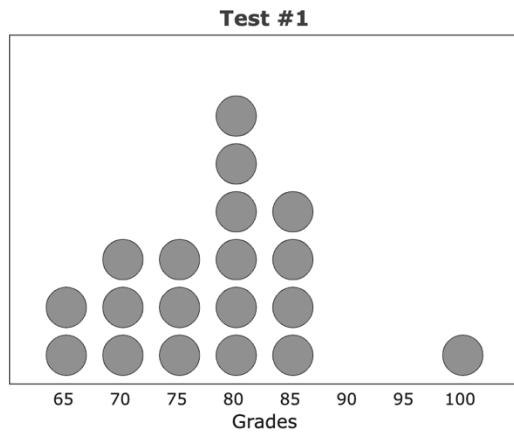
7. The data shown in the table is ☐ univariate.
☐ bivariate.
8. The data in the table should be displayed in a ☐ line plot.
☐ scatter plot.
9. Explain your reasoning from question 8.
10. Create a display to represent the relationship between the number of people in a household and the weekly grocery costs.



Algebra 1
Unit 12
Lesson 3 Practice

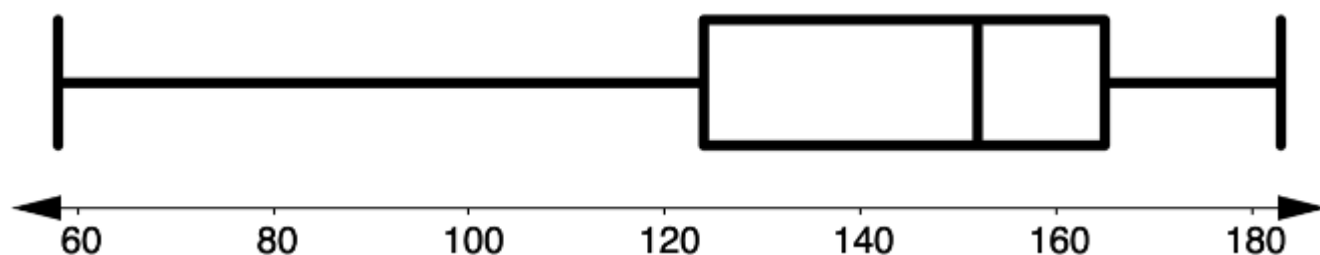
Name _____
Date _____
Period _____

Two dot plots are shown, with results from two recent quizzes shown.



1. What is the mean of the scores for Test #1?
2. What is the median for the scores of Test #1?
3. Which measure of center is the most appropriate to use to describe the scores of Test #1? Explain how you know.
4. Find the interquartile range of Test #1 and interpret it in the context of the problem.
5. What is the mean of the scores for Test #2?

6. What is the median of the scores for Test #2?
7. Which measure of center is most appropriate to use to describe the scores of Test #2?
8. Write a sentence comparing the results of Test #1 to Test #2 using measures of center and measures of spread.
9. In the box plot shown, which measure of center would best be used to describe the data? How do you know?



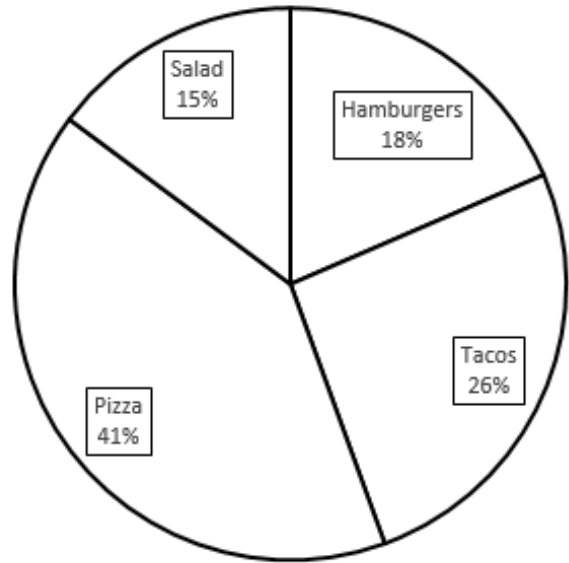
10. A table of values for the heights of a group of tomato plants has a mean of 13 cm and a standard deviation of 0.2 cm. Interpret the variation in height.

Algebra 1
Unit 12
Lesson 4 Practice

Name _____
Date _____
Period _____

A circle graph is shown, with the results of a survey of a classroom of 135 students about which lunch entrée they prefer.

1. What is the most frequently chosen lunch entrée?
2. What is the least frequently chosen lunch entrée?
3. How many students in the class chose tacos?
4. How many students in the class chose salad?
5. How many more students chose pizza over hamburgers?



A table is shown with the results of a survey of a group of 130 students about their favorite type of book.

Favorite Book Genre	Number of Students
Historical Fiction	45
Non-Fiction	15
Science Fiction	20
Fantasy	50

6. Create a bar graph showing the results of this survey.
7. Create a relative frequency table showing the results of this survey.
8. Create a circle graph showing the results of this survey.
9. What percentage of students preferred either science fiction or historical fiction?
10. What percentage of students did not prefer non-fiction?

Lesson 5 Practice

Period_____

A study shows that temperature has a negative correlation with hot chocolate sales.

4. What other information would be helpful to determine whether falling temperatures causes hot chocolate sales to rise?

At a theme park, there is a positive correlation between the temperature increasing and the number of lunch time restaurant reservations.

5. What does it mean that there is a positive correlation?
6. Does this mean that increased temperature leads to more lunch reservations being made?
7. What other factors might contribute to this association?

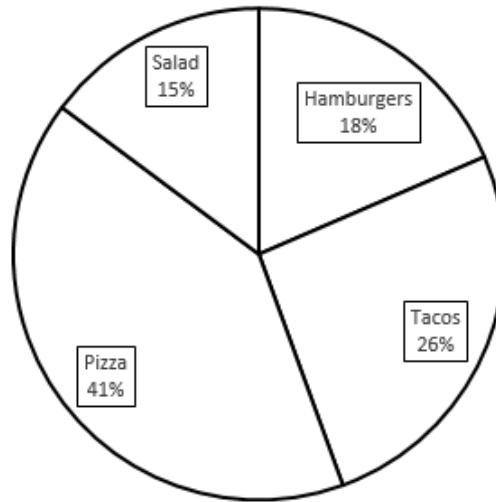
At a water park, as rain increases, attendance decreases.

8. What kind of correlation does this association represent?
9. Is there evidence of a causal relationship?
10. What other information would be helpful to evaluate the claim of a causal relationship?

Algebra 1
Unit 12
Lesson 6 Practice

Name _____
Date _____
Period _____

A circle graph is shown, with the results of a survey of middle school students about which lunch entrée they prefer, with a margin of error of 3%. There are 1250 students in the school.



1. Based on this survey, what is the estimated number of students who would prefer to have tacos?
2. Based on this survey, what is the minimum number of students who would prefer to have hamburgers?
3. Based on this survey, what is the maximum number of students who would prefer to have salad?
4. Based on this survey, what is the range of students who would prefer to have tacos?
5. Based on this survey, what is the range of students who would prefer to have pizza?

A survey of parents indicated that 52% of students felt they have too much homework, with a margin of error of 2.1%.

6. Out of a group of 500 students, how many would likely feel that they have too much homework?

7. Out of a group of 800 students, how many would likely not feel that they have too much homework?

In a survey of 125 families, 24% reported that they eat out less than twice a week, 40% report they eat out twice a week, and 36% report they eat out more than twice a week. The survey has a margin of error of 1.5%.

8. What is the expected number of families that eat out less than twice a week?

9. What is the range of families that eat out twice a week?

10. What is the range of families that eat out less than twice a week?